Coordination-Dependent Hydrogen Adsorption on Co-, Fe-, and Ni-Functionalized Monolayer $CrCl_3$

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Supporting Information Available

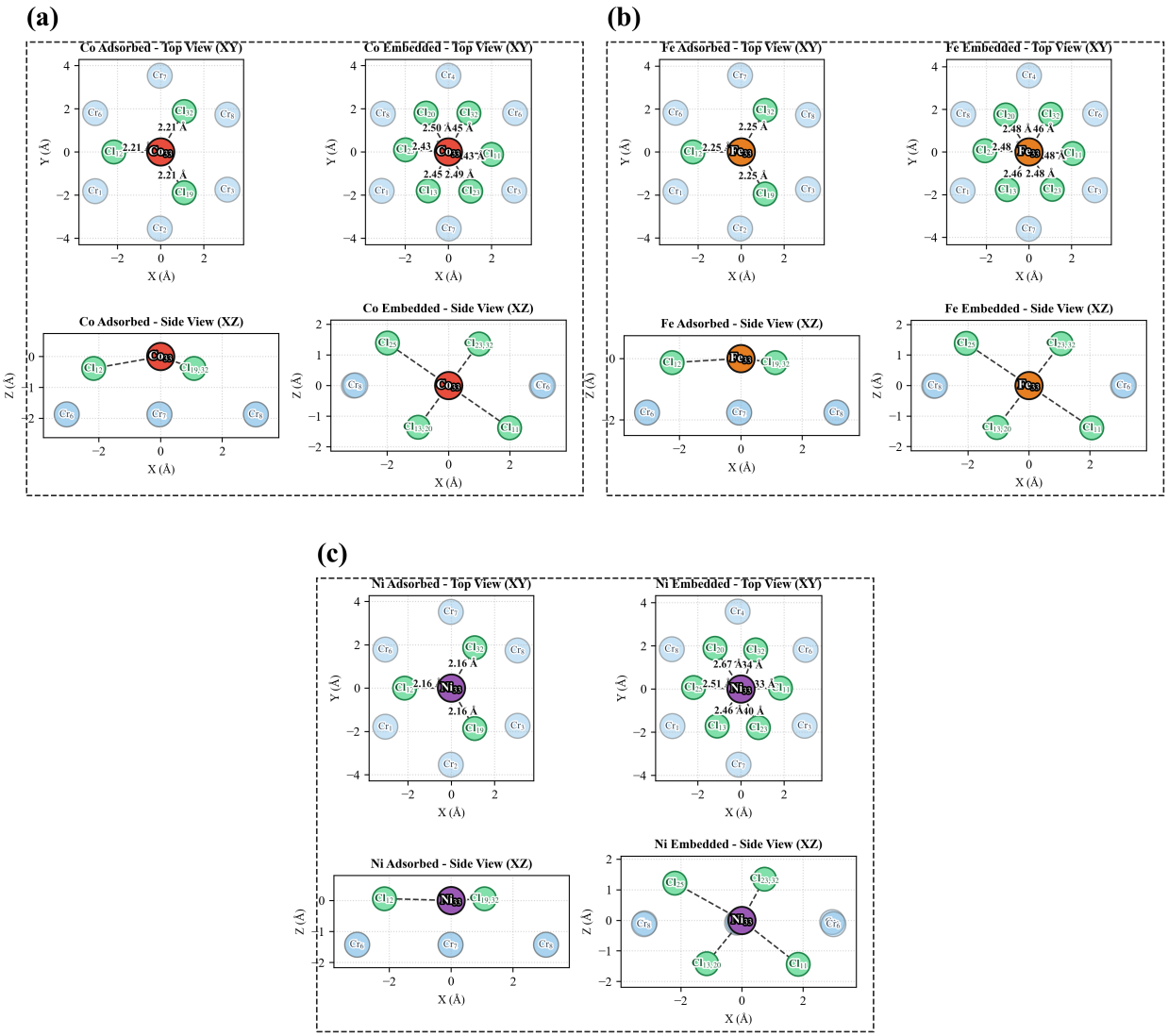


Fig. S1. Schematic top (XY) and side (XZ) views of the optimized local coordination geometries and TM–Cl bond lengths (Å) for (a) Co-, (b) Fe-, and (c) Ni-functionalized monolayer $CrCl_3$ in both adsorbed (hollow site, 3-fold coordinated) and embedded (6-fold octahedrally coordinated) c onfigurations. Light blue, light green, and red/orange/purple spheres represent Cr, Cl, and transition-metal (Co, Fe, Ni) atoms, respectively.

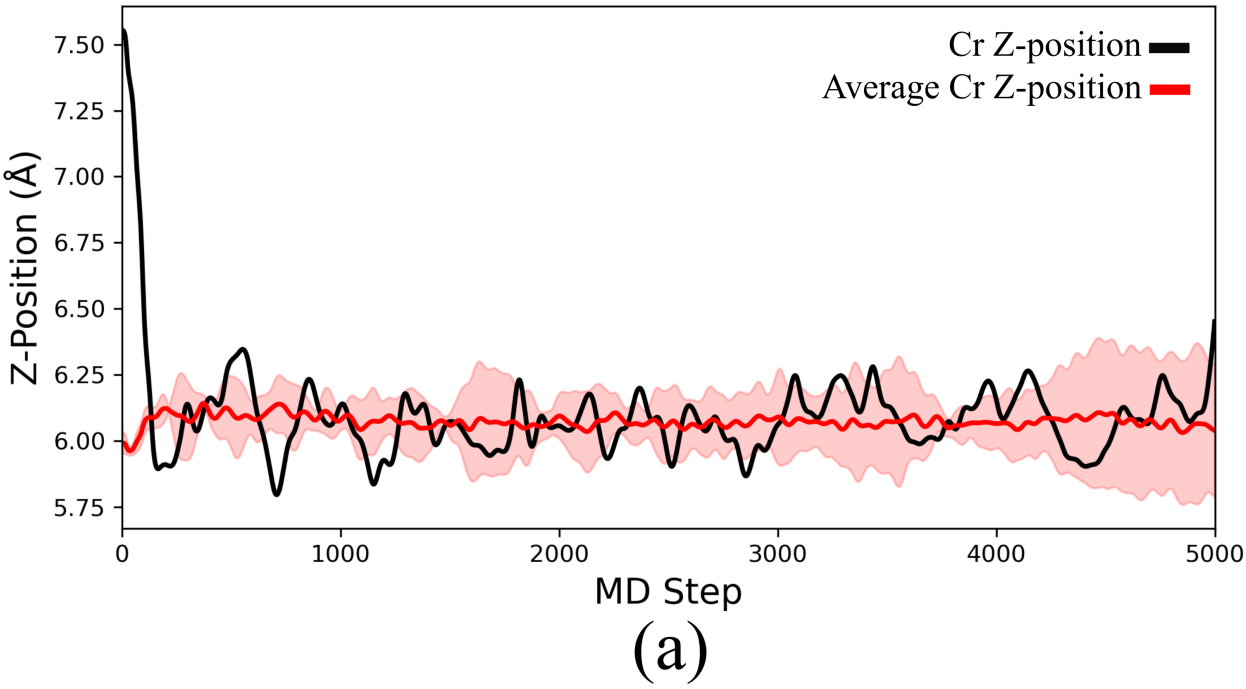


Fig. S2. Time evolution of the absolute z coordinate of the Cr atom initially adsorbed at the hollow site of monolayer $CrCl_3$ during a 5 ps *ab initio* molecular dynamics simulation at 300 K. The black curve represents the instantaneous z coordinate of the adsorbed Cr atom, whereas the red curve denotes the average z coordinate of the Cr atoms belonging to the host monolayer.

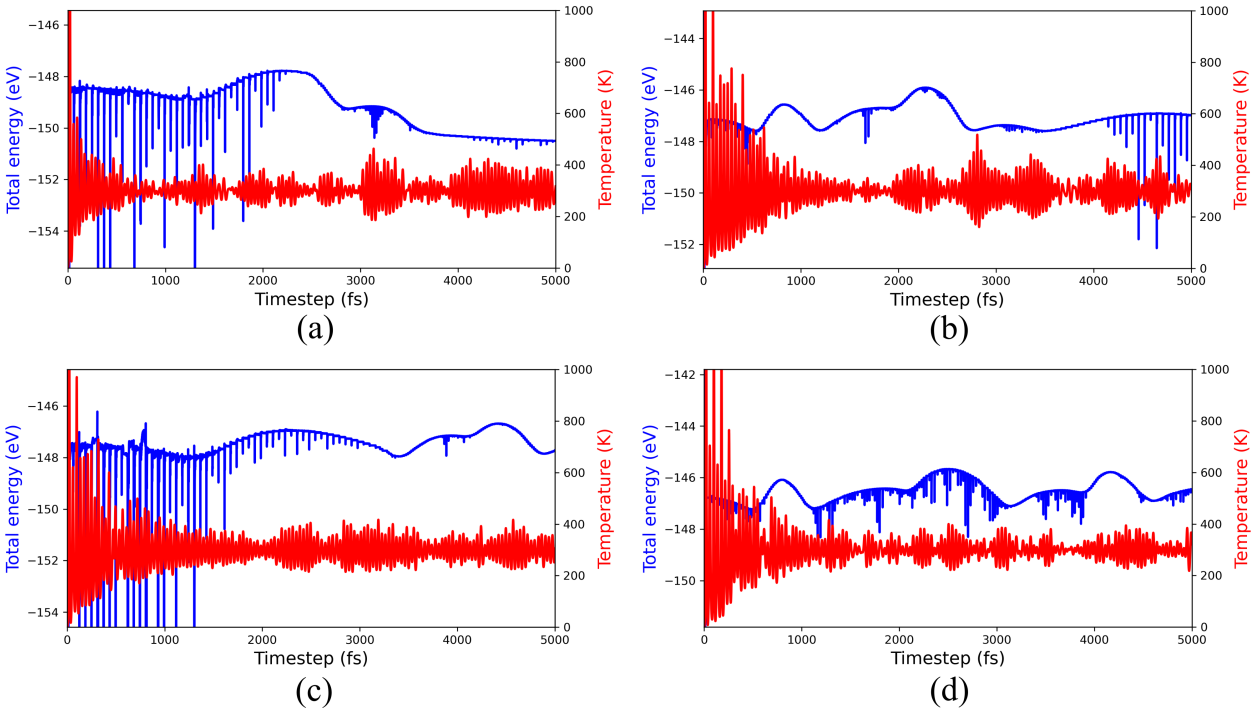


Fig. S3. Time evolution of the total energy (blue curves, left axes) and instantaneous temperature (red curves, right axes) during the 5 ps AIMD simulations at 300 K for monolayer $CrCl_3$ containing an initially surface-adsorbed (a) Cr, (b) Co, (c) Fe, or (d) Ni atom at the hollow site. Following the initial thermostatting transients, the temperatures fluctuate around the target value. The changes in the total-energy profiles accompany the structural rearrangements sampled during the trajectories.

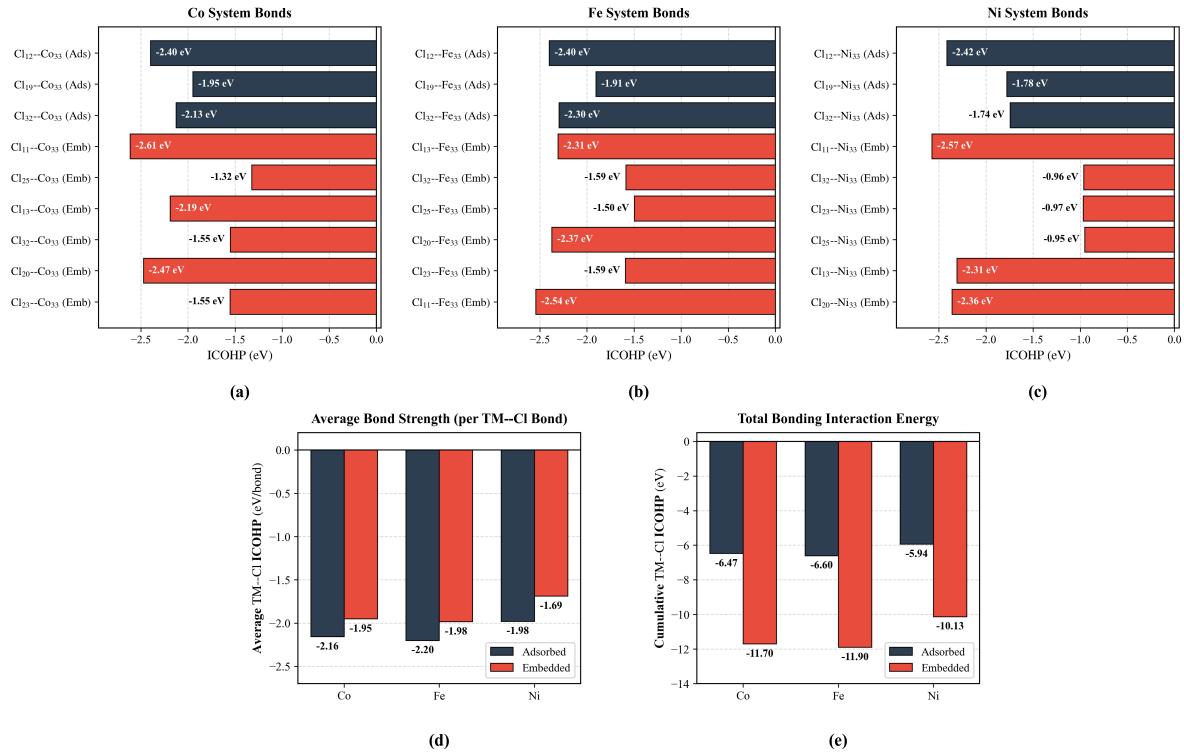


Fig. S4. TM–Cl chemical-bonding analysis and relative stability of surface-adsorbed and embedded transition-metal configurations in monolayer $CrCl_3$. (a–c) Individual integrated crystal orbital Hamilton population (ICOHP) values for the Co–Cl, Fe–Cl, and Ni–Cl interactions, respectively. Dark bars correspond to the threefold coordinated surface-adsorbed configurations, whereas red bars correspond to the sixfold coordinated embedded configurations. More negative ICOHP values indicate stronger net bonding interactions. (d) Average ICOHP per TM–Cl bond for the adsorbed and embedded configurations. (e) Cumulative TM-Cl ICOHP obtained by summing the individual ICOHP contributions over the corresponding coordination shell.

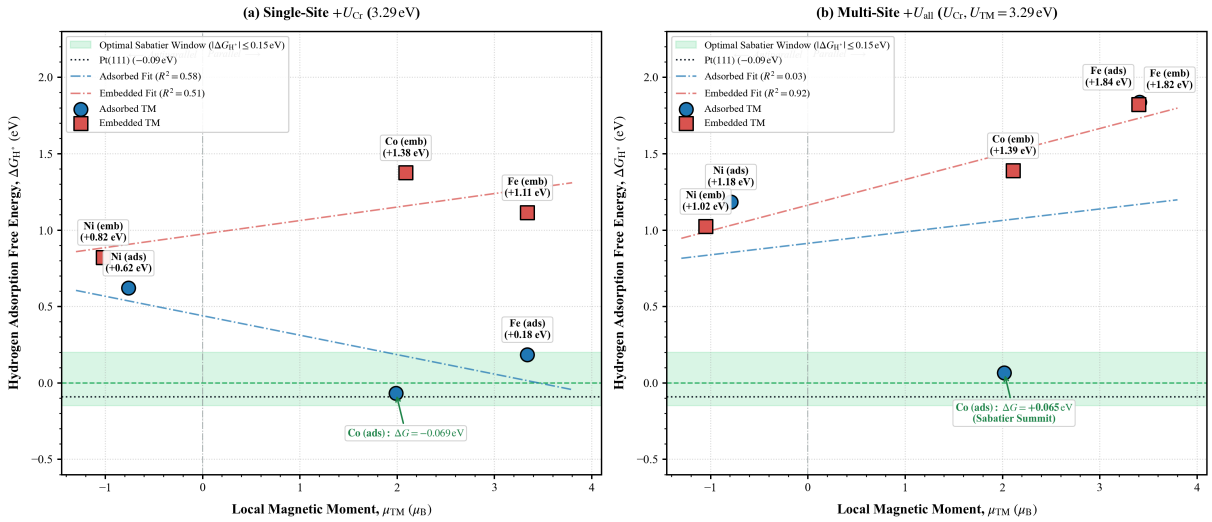


Fig. S5. Relationship between the local magnetic moment of the functionalizing transition-metal atom, μ_{TM}, in the relaxed H-free configurations and the hydrogen adsorption free energy, ΔG_{H^*}, across two systematic Hubbard U tiers: (a) single-site $+U_{\mathrm{Cr}}$ ($U_{\mathrm{Cr}} = 3.29$ eV) and (b) multi-site $+U_{\mathrm{all}}$ ($U_{\mathrm{Cr}} = U_{\mathrm{TM}} = 3.29$ eV). Blue circles and red squares represent the surface-adsorbed (S_2) and embedded transition-metal configurations in monolayer $CrCl_3$, respectively. The vertical dash-dotted line at $\mu_{\mathrm{TM}} = 0$ separates antiparallel (Ni) from parallel (Co, Fe) magnetic coupling relative to the ferromagnetic Cr host. Shaded green bands denote the optimal Sabatier catalytic window ($|\Delta G_{\mathrm{H}^*}| \leq 0.15$ eV) with the Pt(111) benchmark (-0.09 eV) indicated by dotted horizontal lines. Magnetic moments are expressed in units of the Bohr magneton, μ_{B}.